# Nonparametric Models

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## Empirical CDF

**Definition 0.1** (Empirical CDF)**.** For observed values $x_1, \ldots, x_n$, the **empirical cumulative distribution function** (**empirical CDF**) is the function

$$\hat{F}(t) \stackrel{\text{def}}{=} \frac{1}{n} \sum_{j=1}^{n} I(x_j \leq t), \quad t \in \mathbb{R},$$

where $I(A)$ is the indicator function: $I(A) = 1$ if $A$ holds, and $I(A) = 0$ otherwise.

**Example 0.1** (Numerical example: empirical CDF)**.** Suppose the observed values are $4, 1, 7, 3$, so $n = 4$. Two of the four values (1 and 3) are at most 3.5, so:

$$\begin{aligned}
\hat{F}(3.5) &= \frac{1}{4}(I(4 \leq 3.5) + I(1 \leq 3.5) + I(7 \leq 3.5) + I(3 \leq 3.5)) && \text{(definition of } \hat{F}) \\
&= \frac{1}{4}(0 + 1 + 0 + 1) && \text{(evaluate each indicator)} \\
&= 0.5 && \text{(arithmetic)}
\end{aligned}$$

Similarly, $\hat{F}(0) = 0$, because no value is at most 0, and $\hat{F}(7) = 1$, because every value is at most 7.

## Order statistics

**Definition 0.2** (Order statistics)**.** For observed values $x_1, \ldots, x_n$, the **order statistics** are the same values sorted in increasing order, written

$$x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}.$$

So $x_{(1)}$ is the smallest value and $x_{(n)}$ is the largest.

**Example 0.2** (Numerical example: order statistics)**.** If the observed values are $4, 1, 7, 3$, then the order statistics are $x_{(1)} = 1$, $x_{(2)} = 3$, $x_{(3)} = 4$, and $x_{(4)} = 7$.

## Sample quantiles

**Definition 0.3** (Sample quantile function)**.** Given an empirical CDF $\hat{F}$, the **sample quantile function** (also called the **empirical quantile function**) is

$$\hat{Q}(p) \stackrel{\text{def}}{=} \inf\{t : \hat{F}(t) \geq p\}, \quad 0 < p \leq 1.$$

**Definition 0.4** (Sample quantile)**.** For a given $p$, the value $\hat{Q}(p)$ of the sample quantile function is the **sample $p$ quantile**, also called the **sample $100p$th percentile**.

The CDF $F$ and the quantile function $Q$ describe a probability distribution, while $\hat{F}$ and $\hat{Q}$ describe a sample, and serve as estimates of $F$ and $Q$.

**Theorem 0.1** (Order-statistics form of the sample quantile)**.** *If $x_{(1)} \leq \cdots \leq x_{(n)}$ are the order statistics and $i \in \{1, \ldots, n\}$, then $\hat{Q}(p) = x_{(i)}$ for every $p$ with $(i-1)/n < p \leq i/n$. In particular, $\hat{Q}(i/n) = x_{(i)}$.*

Proof

*Proof.* Fix $p$ with $(i-1)/n < p \leq i/n$.
For any $t < x_{(i)}$, the $n - i + 1$ values $x_{(i)}, \ldots, x_{(n)}$ all exceed $t$, so at most $i - 1$ values are at most $t$, and:

$$\begin{aligned} \hat{F}(t) &\leq \frac{i-1}{n} && \text{(at most } i-1 \text{ indicators equal 1)} \\ &< p && \text{(choice of } p\text{)} \end{aligned}$$

So no $t < x_{(i)}$ belongs to the set $\{t : \hat{F}(t) \geq p\}$.
At $t = x_{(i)}$, the $i$ values $x_{(1)}, \ldots, x_{(i)}$ are all at most $x_{(i)}$, so:

$$\begin{aligned} \hat{F}\big(x_{(i)}\big) &\geq \frac{i}{n} \quad \text{(at least } i \text{ indicators equal 1)} \\ &\geq p \quad \text{(choice of } p\text{)} \end{aligned}$$

So $x_{(i)}$ belongs to the set, and it is the smallest member of the set, because no smaller $t$ belongs to it. Therefore $\hat{Q}(p) = x_{(i)}$. □

**Example 0.3** (Numerical example: sample quantiles)**.** Using the same data $4, 1, 7, 3$, with order statistics $1, 3, 4, 7$, Theorem 0.1 says the sample quantile function is a step function:

- $\hat{Q}(p) = 1$ for $0 < p \leq 1/4$;
- $\hat{Q}(p) = 3$ for $1/4 < p \leq 1/2$;
- $\hat{Q}(p) = 4$ for $1/2 < p \leq 3/4$;
- $\hat{Q}(p) = 7$ for $3/4 < p \leq 1$.

So $\hat{Q}(0.5) = 3$ and $\hat{Q}(0.9) = 7$. R's `quantile()` function computes this definition when called with `type = 1`:

```
quantile(c(4, 1, 7, 3), probs = c(0.5, 0.9), type = 1)
#> 50% 90%
#>   3   7
```

*Remark* 0.1 (Quantile conventions). Other sources, and other software defaults, define sample quantiles differently, mostly by interpolating between order statistics. R's `quantile()` offers nine definitions through its `type` argument, and its default (`type = 7`) interpolates, so `quantile(c(4, 1, 7, 3), 0.5)` returns 3.5, not 3. The usual sample median is another interpolated quantile: for an even number of observations, it averages the two middle order statistics. These notes use Definition 0.3, which always returns one of the observed values.

### The empirical CDF and quantile function as generalized inverses

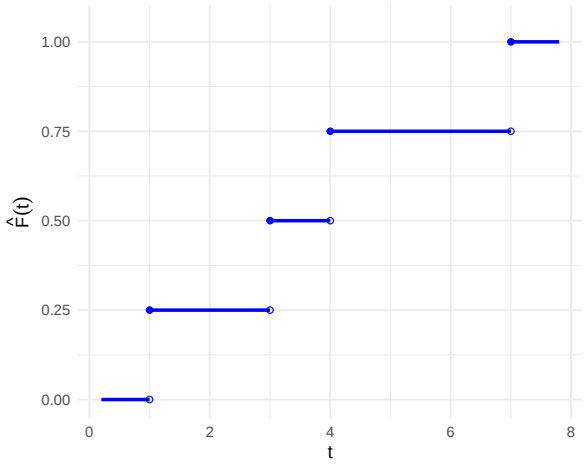
Both functions in Figure 1 are step functions, so neither is one-to-one. Each flat piece of one function corresponds to a jump of the other: for example, $\hat{F}(t) = 0.5$ for every $t \in [3, 4)$, and $\hat{Q}$ jumps from 3 to 4 at $p = 0.5$. So $\hat{Q}$ is a *generalized* inverse of $\hat{F}$, not an inverse in the usual sense: $\hat{Q}(\hat{F}(t)) \leq t$ for every $t \geq x_{(1)}$, with equality only at the observed values.

```r
x <- c(4, 1, 7, 3)
n <- length(x)
x_ord <- sort(x)
p_ord <- seq_len(n) / n
ecdf_x_padding <- 0.8
x_min <- min(x_ord) - ecdf_x_padding
x_max <- max(x_ord) + ecdf_x_padding

x_left <- c(x_min, x_ord)
x_right <- c(x_ord, x_max)
f_levels <- c(0, p_ord)

ecdf_pieces <- tibble::tibble(x = x_le
ecdf_open <- tibble::tibble(x = x_ord,
ecdf_closed <- tibble::tibble(x = x_or

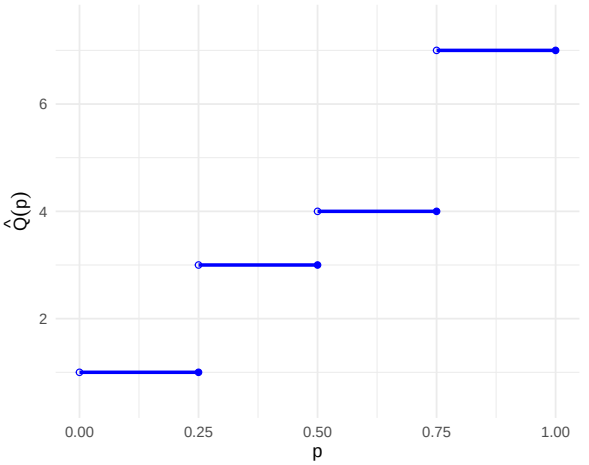
ggplot2::ggplot() +
  ggplot2::geom_segment(
    ggplot2::aes(x = x, xend = xend, y
    data = ecdf_pieces, linewidth = 1,
  ) +
  ggplot2::geom_point(
    ggplot2::aes(x = x, y = y), data =
  ) +
  ggplot2::geom_point(
    ggplot2::aes(x = x, y = y), data =
  ) +
  ggplot2::coord_cartesian(xlim = c(x_m
  ggplot2::labs(x = "t", y = expression
  ggplot2::theme_minimal()
```




(a) Empirical CDF, horizontal pieces only. Closed circles mark included endpoints; open circles mark excluded endpoints.

```r
eqf_y_padding <- 0.5
q_left <- c(0, p_ord[-n])
q_right <- p_ord
eqf_pieces <- tibble::tibble(x = q_left, xend = q_righ

ggplot2::ggplot(eqf_pieces) +
  ggplot2::geom_segment(
    ggplot2::aes(x = x, xend = xend, y = y, yend = y),
    linewidth = 1, color = "blue"
  ) +
  ggplot2::geom_point(ggplot2::aes(x = x, y = y), shap
  ggplot2::geom_point(
    ggplot2::aes(x = xend, y = y), shape = 19, color =
  ) +
  ggplot2::coord_cartesian(
    xlim = c(0, 1),
    ylim = c(min(x_ord) - eqf_y_padding, max(x_ord) +
  ) +
  ggplot2::labs(x = "p", y = expression(hat(Q)(p))) +
  ggplot2::theme_minimal()
```




(b) Sample quantile function, horizontal pieces only. Open circles mark excluded left endpoints; closed circles mark included right endpoints.

Figure 1: The empirical CDF and the sample quantile function for the data 4, 1, 7, 3 (order statistics 1, 3, 4, 7).